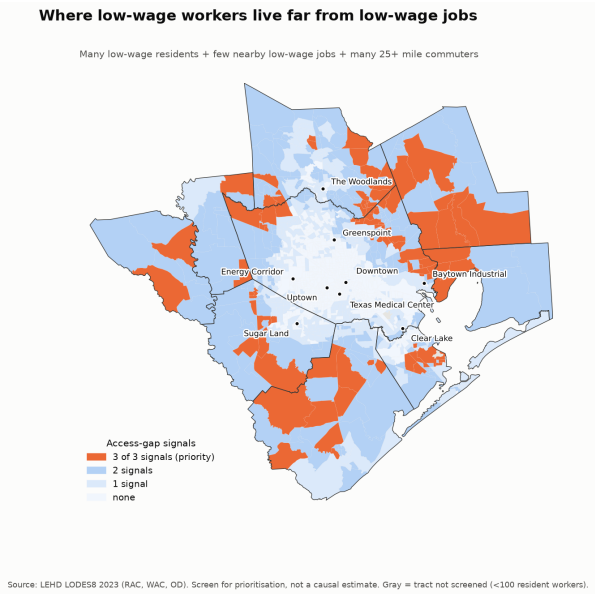
Houston Employment & Commute Analysis

Where do people in Houston work, where do they live, and who is left far away from jobs?

Data analytics portfolio project using US Census LEHD LODES8 data, 2010-2023

Tools: Python, SQL (SQLite), geospatial mapping



Priority neighbourhoods found by the analysis

Repository: github.com/Balavardhanreddy5872/TX-Employment-Trends

1. Project overview

The question. Where should the Houston region spend money to help people reach jobs, given how jobs and commutes have changed since 2010?

The idea in simple words. Every worker lives in one place and works in another. The Census Bureau keeps an anonymous record of this for every worker in the country. I used that record for Texas to see where jobs are growing, where workers live, and how far they travel. Then I looked for neighbourhoods where low-wage workers live far away from low-wage jobs.

The answer in short.

- Houston grew **more slowly** than Dallas-Fort Worth, Austin and San Antonio after the 2015 oil price crash.
- New jobs are appearing in the **suburbs** (Montgomery, Fort Bend counties), not in downtown Houston.
- As a result, commutes are getting **longer**.
- About **50,000 low-wage workers** in **93 neighbourhoods** live far from low-wage jobs. These are the best places to start helping.

Item	Detail
Data source	US Census Bureau, LEHD LODES version 8 (Texas). Free and public.
Years covered	2010 to 2023 (14 years)
Data size	About 450 MB of raw files, 31 files in total, over 12 million Texas jobs
Study area	Houston-The Woodlands-Sugar Land region: 9 counties, 1,606 census tracts
Compared against	Dallas-Fort Worth, Austin, San Antonio and the Texas average
Tools	Python (pandas, geopandas, matplotlib, folium), SQL (SQLite), pytest, Makefile

2. The data

LODES is built from state unemployment-insurance records. It describes jobs at the level of a census **block** (roughly a city block). Three main files were used:

File	What it tells you	Simple example
WAC (Workplace Area Characteristics)	How many jobs are located in each place, split by industry, wage and age	"There are 5,000 jobs in this downtown tract"
RAC (Residence Area Characteristics)	How many workers live in each place, with the same splits	"2,000 workers live in this suburb"
OD (Origin-Destination)	How many workers travel from each home block to each work block	"300 people go from this suburb to downtown"
Crosswalk and boundaries	Links blocks to tracts, counties and metro areas, plus map shapes	Needed to group data and draw maps

Why this data is useful: WAC says *where jobs are*, RAC says *where people live*, and OD joins the two to show *how far people travel*. Comparing them reveals places that have many workers but few jobs nearby.

Useful detail: LODES8 uses the same 2020 census blocks for every year, so a neighbourhood means the same thing in 2010 and in 2023. No re-matching of boundaries was needed.

Job type used: all jobs (JT00) and all workers (S000). The OD file was used for 2015, 2019 and 2023 because it is very large.

3. Step-by-step workflow



Step 1 - Download the data

A Python script (`src/download_data.py`) downloads all files from the Census website with a progress bar. It is **idempotent**: if a file already exists it is skipped, so you can re-run it safely.

Step 2 - Clean and check the data (Python)

The script `src/clean.py` reads each file and runs automatic quality checks. Every file is checked for:

- duplicate rows and wrong ID lengths (every block ID must be 15 characters; IDs are kept as text so leading zeros are not lost),
- negative or missing values,
- totals that do not add up (jobs by industry, by wage and by age must each equal the total jobs),
- blocks that cannot be matched to a tract.

All 31 files passed every check (0 failures). The results are saved in `outputs/reports/data_quality.md`. Then the thousands of small blocks were added up into **tracts** (larger neighbourhoods). Roughly 430,000 residence blocks became 6,900 tracts, more than 98% fewer rows, with total job counts unchanged. The reason: block numbers contain deliberate privacy noise, and tracts are steadier and easier to map.

Step 3 - Store the data in a database (SQLite)

The clean tables were loaded into a **star schema**, a standard way to organise data for analysis:

Table type	Tables	Purpose
Dimension tables (descriptions)	dim_tract, dim_sector, dim_metro, dim_center	Names and attributes of places, industries, metro areas and job centres
Fact tables (numbers)	fact_wac, fact_rac, fact_wac_sector, fact_od	Jobs, workers and commute flows by tract and year

Indexes were added on the join keys. One slow query (294 seconds) was fixed by copying the county code onto the commute table and adding an index, which brought it down to 0.5 seconds.

Step 4 - Analyse with SQL

Six SQL files hold 22 exported queries, one set per business question:

File	Question it answers
01_growth_benchmarks.sql	How did Houston's job growth compare with other Texas metros?
02_houston_structure.sql	Which industries grew or shrank? Is the economy diversified?
03_job_centers.sql	Which job centres (Downtown, Woodlands, Sugar Land...) are growing?
04_commute_flows.sql	How far do people travel, and between which counties?
05_spatial_mismatch.sql	Which neighbourhoods have low-wage workers far from low-wage jobs?

SQL techniques used: CTEs, window functions (LAG, FIRST_VALUE, NTILE, ROW_NUMBER, SUM OVER), pivots with conditional aggregation, self-joins, and a spatial self-join using a custom distance function.

Step 5 - Visualise (Python)

`src/visualize.py` draws 11 colour-blind-safe charts and maps with matplotlib and geopandas, and an interactive tract-level map with folium.

Step 6 - Recommend

The findings were turned into a short policy brief with recommendations, a success measure for each, and a list of limitations. A Makefile (`make all`) rebuilds everything from scratch, and 11 automatic tests (pytest) protect the pipeline.

4. Methodology: key measures explained simply

Measure	Plain-language meaning	How it is calculated
CAGR	Average yearly growth rate over a period	(end / start) to the power of 1/years, minus 1
Location quotient (LQ)	Is an industry more common here than in Texas overall? Above 1 means yes (Houston mining = 1.45)	Industry share of Houston jobs / industry share of Texas jobs
HHI (concentration index)	Is the economy dependent on a few industries? Lower is more diversified	Sum of squared industry shares (in %)
Jobs per resident worker	Above 1 means the area brings in workers from elsewhere; below 1 means workers leave	Jobs in area (WAC) / workers living in area (RAC)
Employment centre	A cluster of jobs such as Downtown or The Woodlands	Tracts whose centre is within 3 miles of a chosen anchor point
Job-worker ratio within 10 miles	How many low-wage jobs exist near each low-wage worker. Low means poor access	Low-wage jobs in tracts within 10 miles / low-wage workers in those tracts
Commute 25+ miles	Share of workers whose job is far from home	OD flows with straight-line distance of 25 miles or more

The access-gap screen (how the 93 priority tracts were found)

Each tract gets one point for each of three warning signs, based on three *different* data files:

Signal	Data used	A tract scores a point if it is in the...
1. Many low-wage workers live here	RAC	top 25% of tracts for low-wage resident workers
2. Few low-wage jobs nearby	WAC and RAC	bottom 25% for low-wage jobs per worker within 10 miles
3. Long commutes	OD	top 25% for low-wage workers travelling 25+ miles

A tract with **all 3 points** is a **priority tract**. Using three independent sources makes the result more trustworthy, because one noisy file cannot decide the outcome. Tracts with fewer than 100 resident workers were not screened.

Study area and comparison

The region is the nine-county Houston-The Woodlands-Sugar Land area (Austin, Brazoria, Chambers, Fort Bend, Galveston, Harris, Liberty, Montgomery and Waller counties). Benchmarks are Dallas-Fort Worth-Arlington, Austin-Round Rock and San Antonio-New Braunfels.

Decisions that improved the analysis

- I dropped a result showing low-wage jobs near workers falling 15%. Wage bands are fixed in old dollars, so inflation pushes jobs into higher bands and fakes a decline. Wage comparisons are now made across places in one year only.
- I replaced average commute distance with the share of 25+ mile commutes, because a few very long or remote-work trips skew the average.
- I used a job-to-worker ratio instead of raw job counts so dense areas are not flagged just for being dense.

5. Findings with charts

Finding 1: Houston fell behind after the oil bust

Houston grew 2.4% a year in 2010-19, in line with Texas (2.3%). From 2019 to 2023 it grew only 1.7% a year, against 2.9% for Dallas-Fort Worth and 4.2% for Austin. Houston added 215,000 jobs since 2019 (+6.9%), while Dallas-Fort Worth added 456,000 (+12.3%).

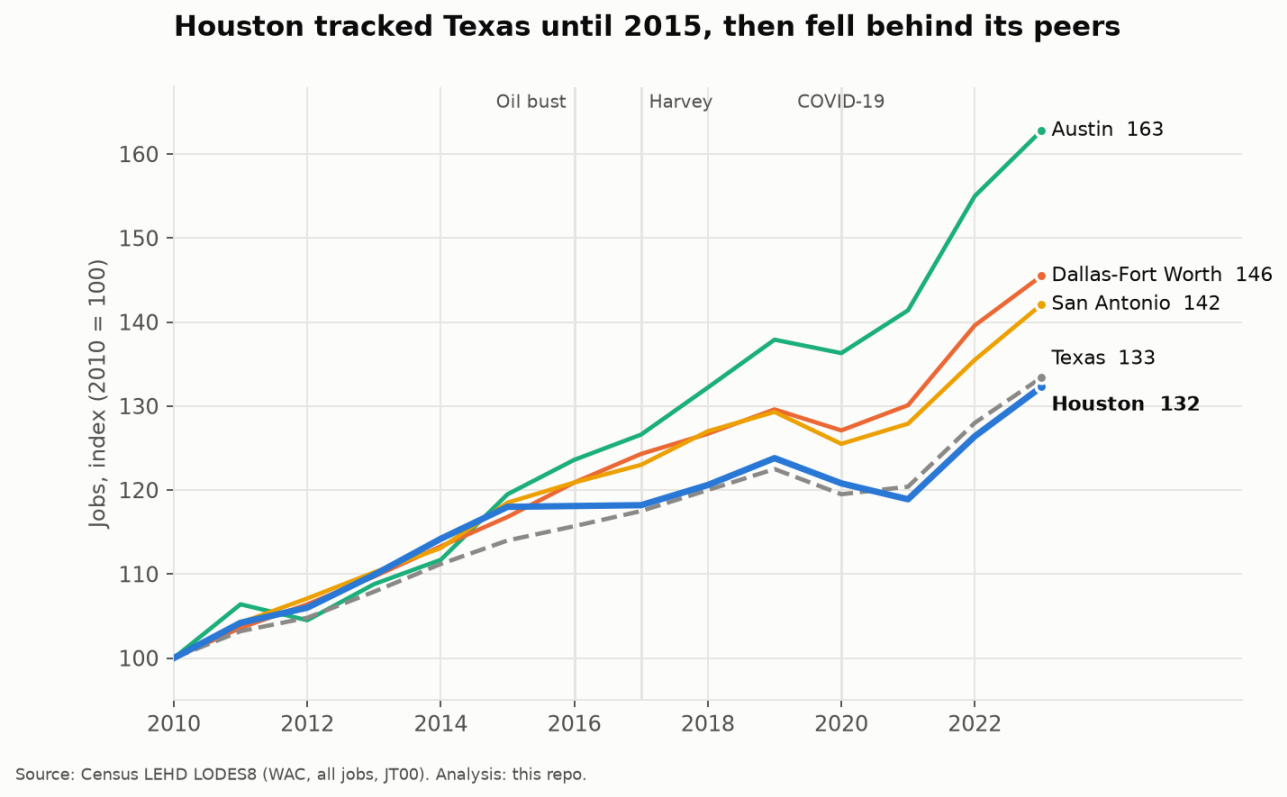


Figure 1. Job growth index by metro area. Houston's line flattens after 2015 and falls below its peers.

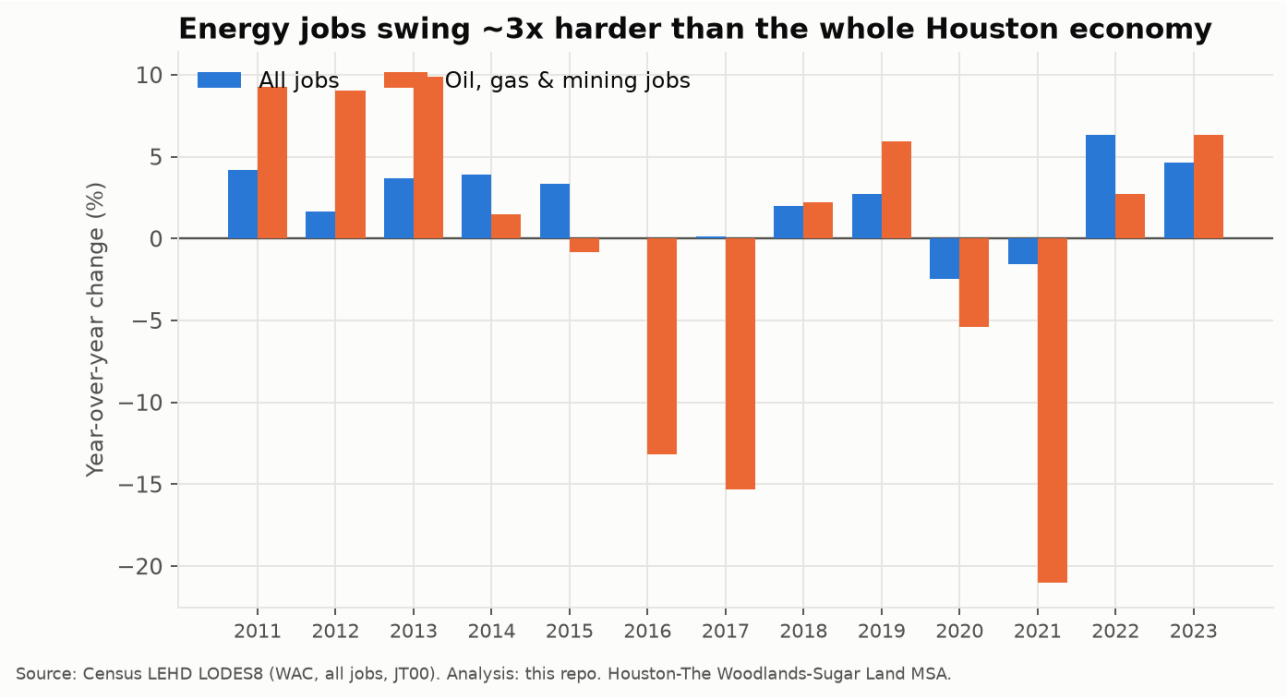


Figure 2. Year-over-year change: energy jobs swing about three times harder than total jobs. In 2016-17 total growth was about zero while mining fell 13-15% a year.

Finding 2: The economy is diversifying, and that is healthy

Oil, gas and mining jobs fell 35% from their 2014 peak (116,000 to 75,000), yet the concentration index stayed about 760, close to Dallas-Fort Worth at 749. Growth since 2019 came from professional and technical services (+42,000), health care (+37,000) and transportation and warehousing (+27,000).

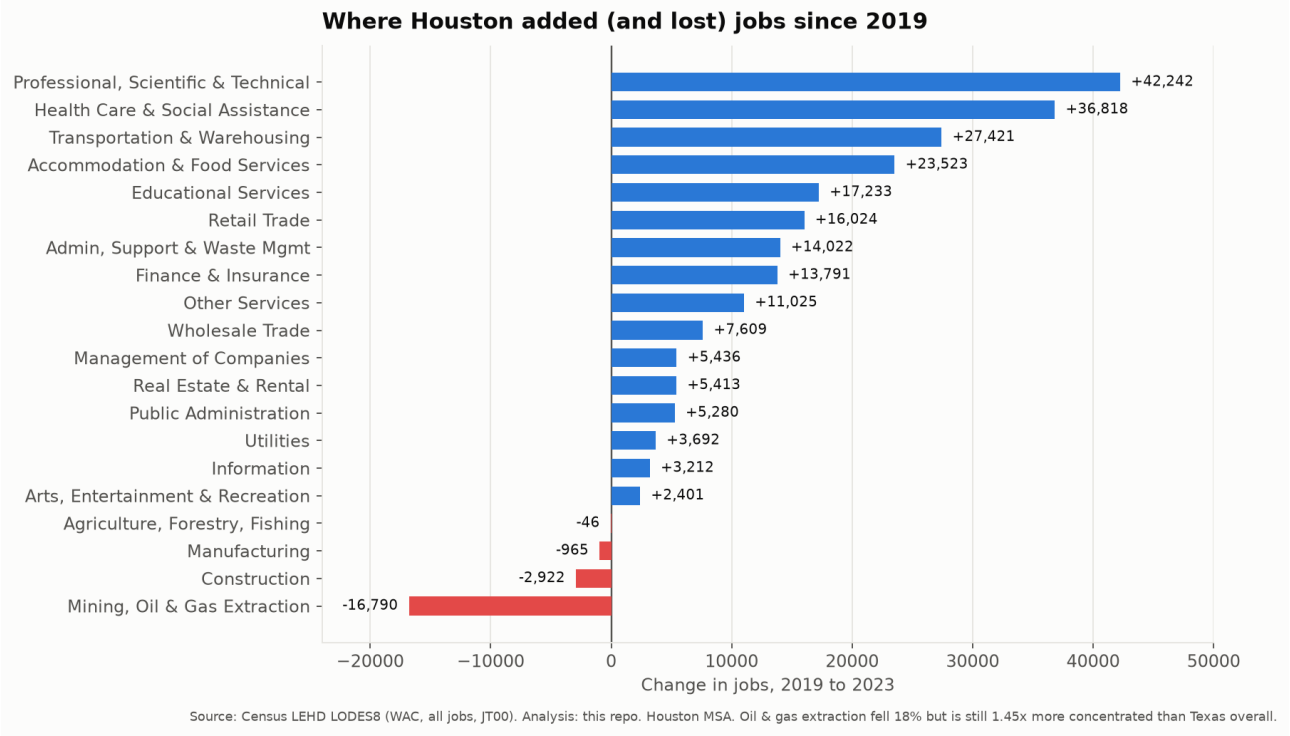


Figure 3. Change in jobs by industry. Professional services, health and logistics lead; mining declines.

Finding 3: Growth is moving to the suburbs

Harris County still holds 76% of the region's jobs, but it grew only 4.5% since 2019. Montgomery grew 20.5%, Fort Bend 16.1% and Chambers 24.6%. Harris has 1.22 jobs per resident worker (it brings workers in), while Fort Bend has 0.60 (its workers leave to work).

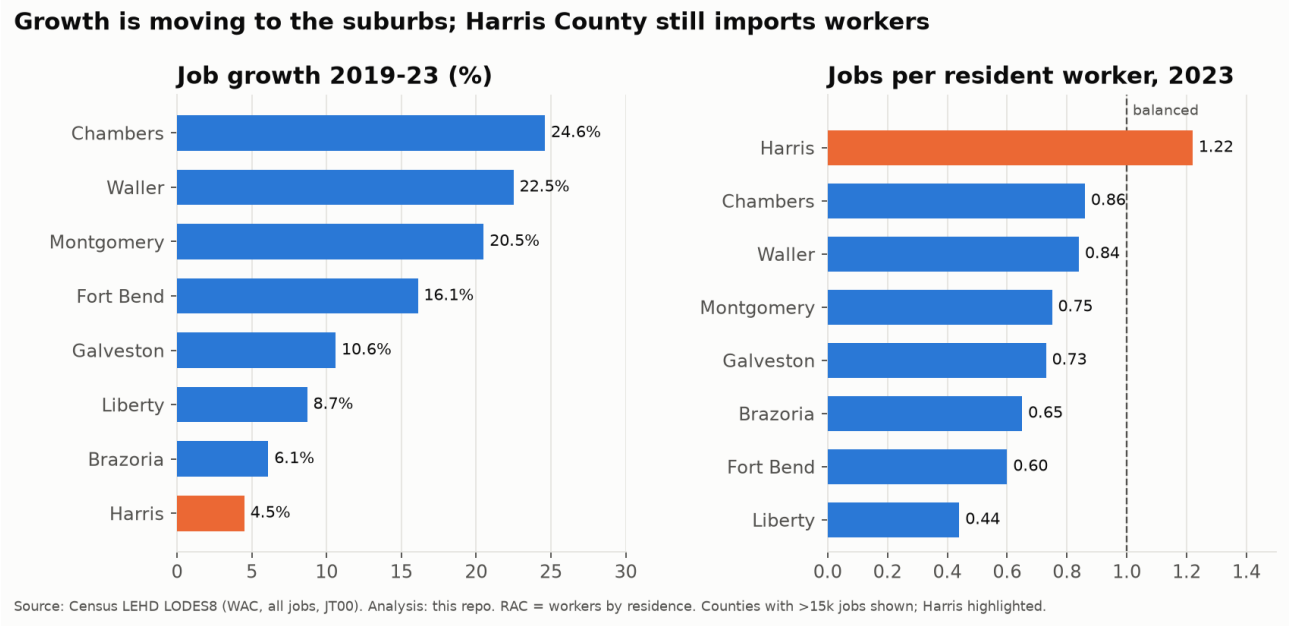


Figure 4. Job growth and jobs-per-worker balance by county.

Finding 4: Job centres are shifting outward

The Woodlands grew 71% since 2010 and Sugar Land 56%, against only 7% for Downtown and 1% for Uptown/Galleria. The share of regional jobs 20-35 miles from Downtown rose from 15.4% to 22.1%, while the inner rings fell from 39.8% to 33.0%.

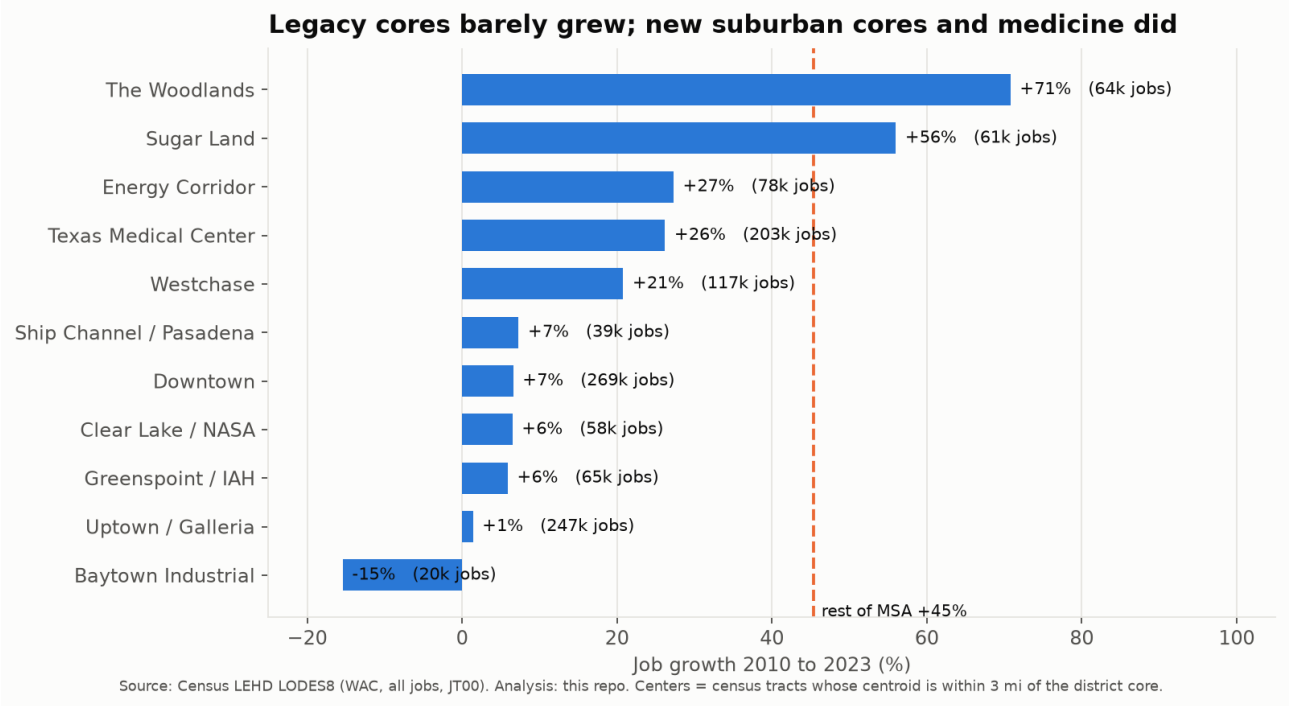


Figure 5. Growth of job centres since 2010. Suburban centres grow fastest.

Finding 5: Commutes are longer and more regional

The share of workers with a job 25+ miles from home rose from 22.5% to 26.9% (2015-23). 37% of Harris County jobs are now held by people who live outside the county (32.5% in 2015). Note: remote workers are counted at the employer's address, so the size of the rise is an upper bound; the direction is reliable.

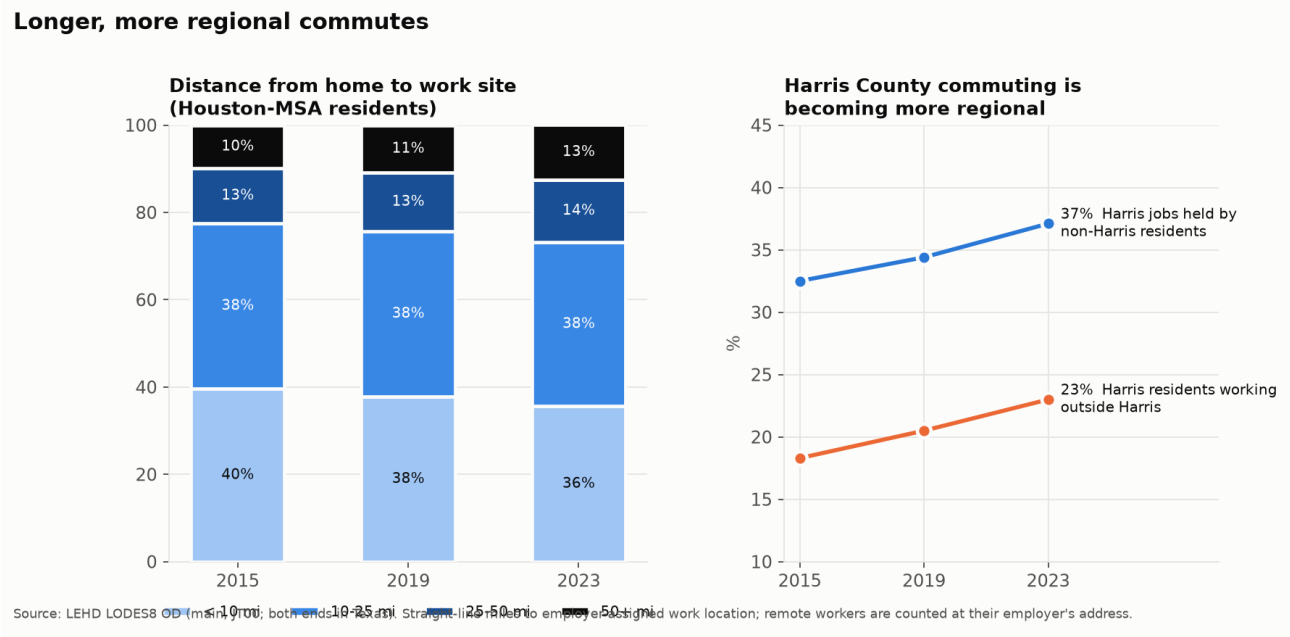


Figure 6. Commute distance trend and commuting into and out of Harris County.

Finding 6: 93 tracts where low-wage workers are stranded

The access-gap screen found 93 priority tracts with 49,704 low-wage workers (9.8% of the region's 505,000). Low-wage jobs within 10 miles per worker: Harris 1.15, Brazoria 0.70, Liberty 0.48. Residents commuting 25+ miles: Harris 20%, Montgomery 47%, Liberty 70%.

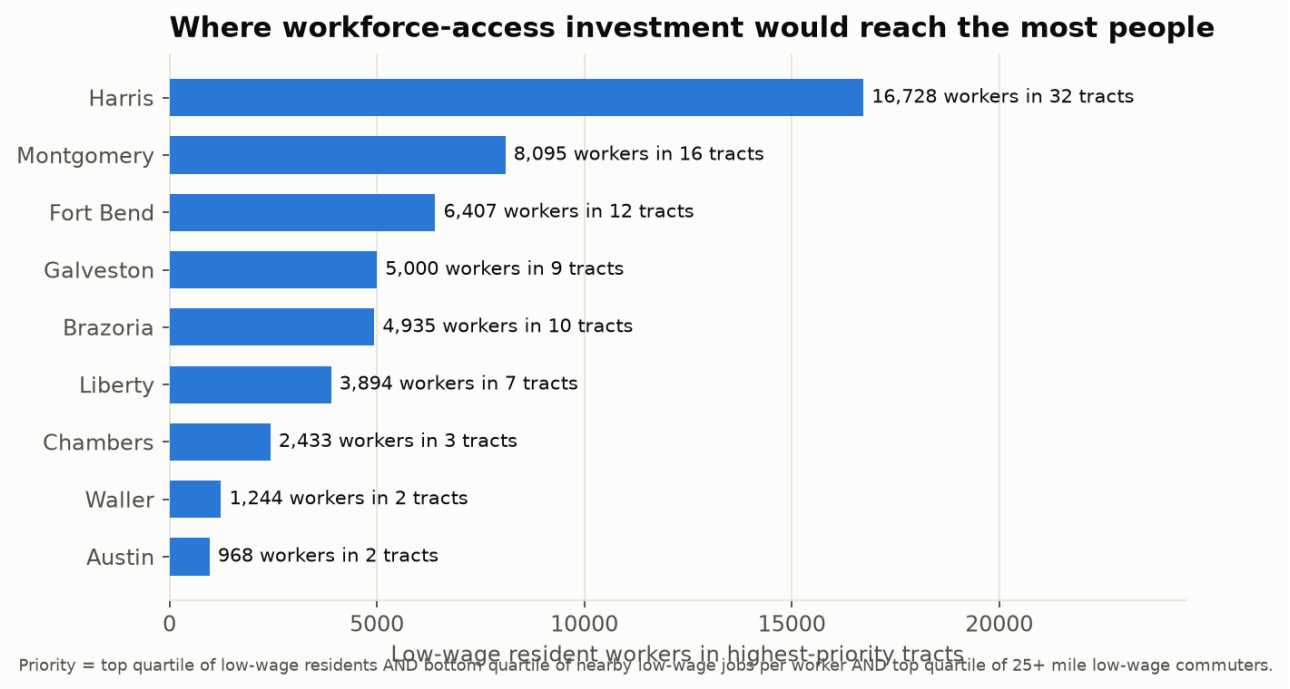


Figure 7. Priority tracts and low-wage workers by county: Harris, Montgomery, Fort Bend, Galveston, Brazoria and Liberty.

County	Priority tracts	Low-wage workers
Harris	32	16,700
Montgomery	16	8,100
Fort Bend	12	6,400
Galveston	9	5,000
Brazoria	10	4,900
Liberty	7	3,900

6. Maps

The maps show the same story geographically. Each shape is a census tract.

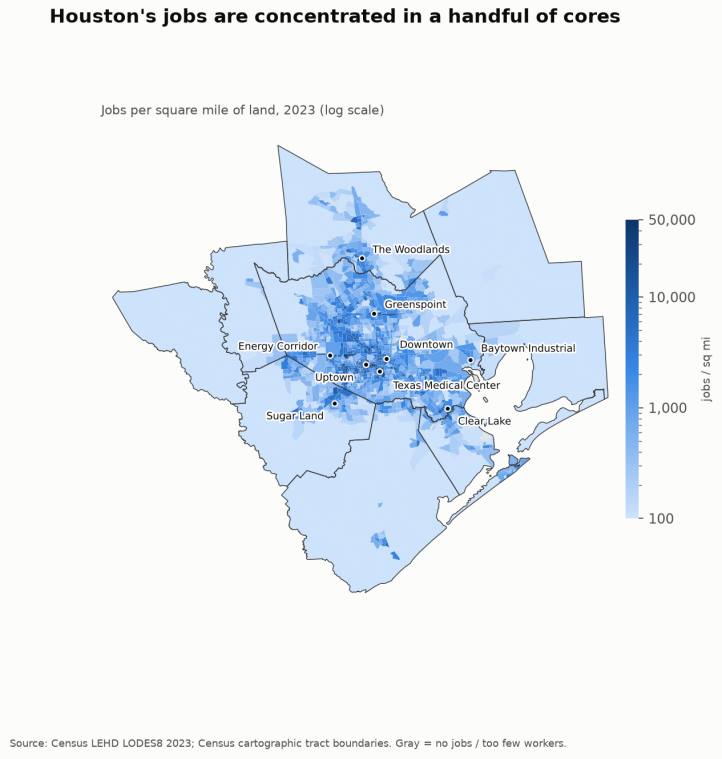


Figure 8. Job density in 2023. Jobs are concentrated in a handful of cores.

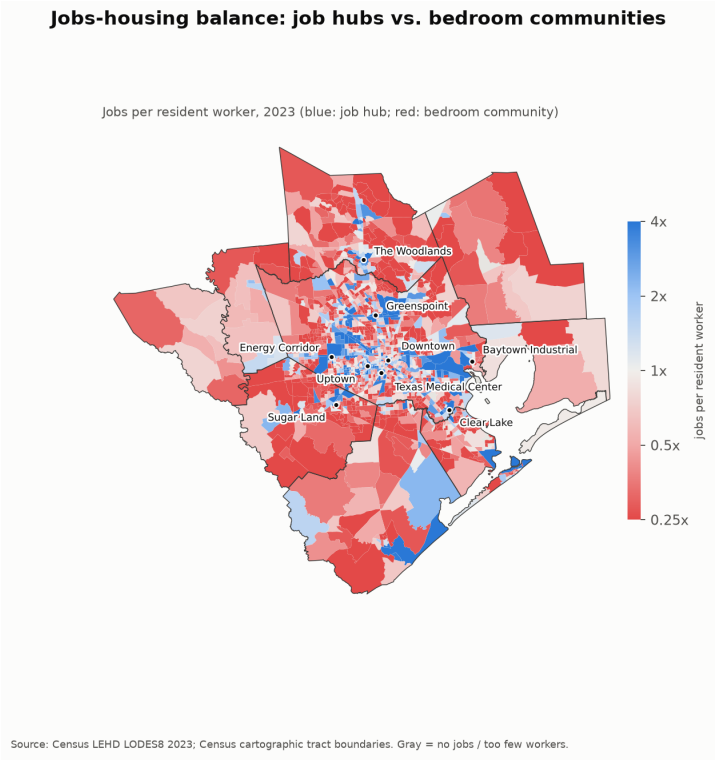


Figure 9. Jobs-housing balance. Areas with many more jobs than resident workers versus areas where workers live but jobs are elsewhere.

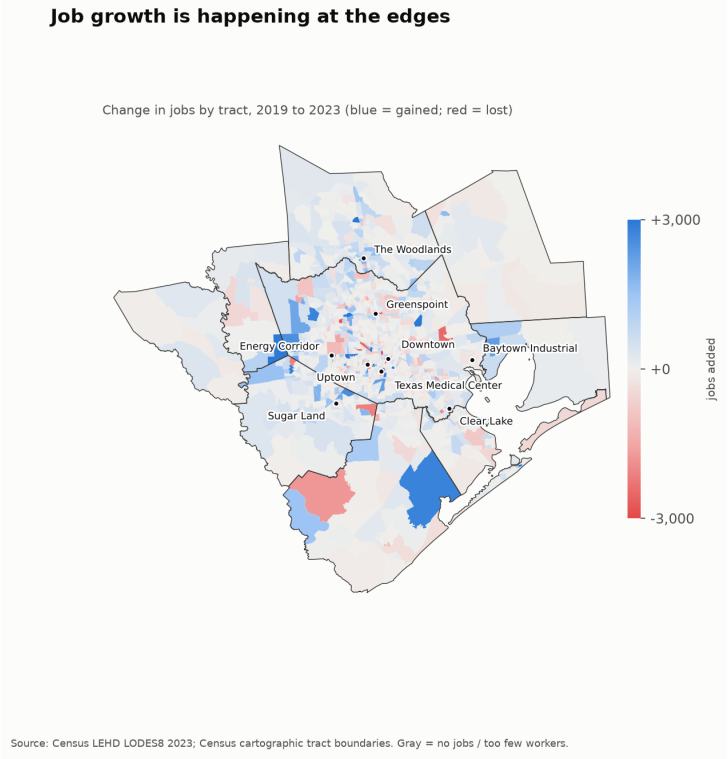


Figure 10. Change in jobs, 2019 to 2023. Job growth is happening at the edges.

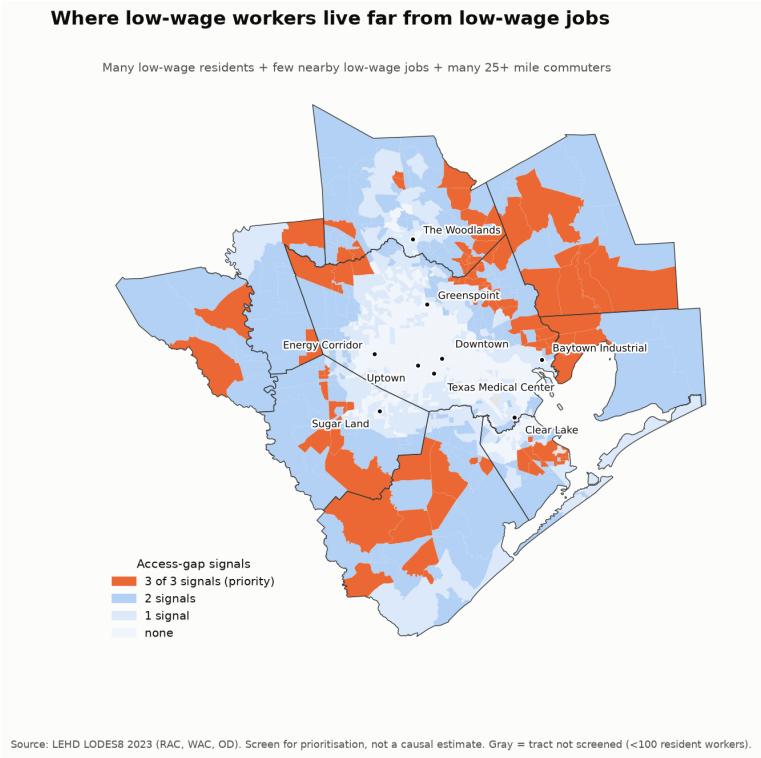


Figure 11. The 93 priority tracts, where all three warning signs appear together.

An interactive version with toggle layers and hover details is in `outputs/maps/houston_jobs_interactive.html` (open it in a web browser).

7. Recommendations

1. Pilot commuter access where the gap is widest. Start with corridors linking priority tracts in Montgomery, Liberty/Chambers and Brazoria to their nearest big job centres (The Woodlands/Greenspoint, Baytown/Ship Channel, Clear Lake/Texas Medical Center). Use vanpools and microtransit first, because long trips from scattered homes are too costly for fixed bus routes. *Success measure: share of low-wage residents in pilot tracts who can reach 25,000+ jobs within 45 minutes.*

2. Anchor suburban job growth in sectors that already work outside the core. Health care, logistics and professional services drove growth. Where low-wage jobs per worker within 10 miles is below about 0.8 (Brazoria, Liberty), use incentives and hiring partnerships to bring jobs to workers instead of the reverse.

3. Treat energy as a cycle, not a trend. Mining is still 1.45 times over-represented and swings far harder than the whole economy. Keep funding diversification and track the mining location quotient and the 20-35 mile job ring every year as early-warning signals.

4. Validate before spending. This is a screening tool. Check vehicle access (ACS), transit routes (GTFS) and employer locations in the 93 priority tracts before committing budget.

8. Limitations

Limitation	How it was handled
Wage bands are in fixed old dollars, so inflation moves jobs between bands	Wages compared across places within one year only, never as a trend over time
Remote workers are counted at the employer's address	Distances reported in bands; the rise in long commutes is treated as direction, not exact miles
Block-level numbers contain privacy noise	All analysis done at tract level or higher
OD data covers only in-state jobs covered by unemployment insurance	Stated in the report; out-of-state flows and self-employed are excluded
Distances are straight-line, not road distance	Used only as relative measures
The three signals are related to each other, and the screen is not causal	Used to decide where to look first, not to prove cause; validation plan included

9. Skills demonstrated

Area	What was done
Python	pandas, numpy, geopandas, matplotlib, folium; modular src/ package; pytest tests
SQL	CTEs, window functions, pivots, self-joins, spatial joins, indexing, query tuning (294 s down to 0.5 s)
Data engineering	Repeatable download-clean-load pipeline, Makefile, star schema, automated data-quality audit of 31 files
Analytics	Growth rates (CAGR), location quotient, HHI, shock analysis (oil bust, Hurricane Harvey, COVID), accessibility measure, quartile screening
Geospatial	Choropleth maps, distance calculations, interactive map
Communication	Policy brief with recommendations and caveats; colour-blind-safe charts
Judgment	Dropped a misleading result, replaced weak metrics, stated limits openly

10. How to explain this project

30-second version

"I used 14 years of Census data to study jobs and commutes in the Houston region. Houston grew slower than other big Texas cities after the oil crash, new jobs moved to the suburbs, and commutes got longer. I built the pipeline in Python and SQL, and found about 93 neighbourhoods where roughly 50,000 low-wage workers live far from low-wage jobs. I recommend piloting commuter access there first."

Numbers worth remembering

Fact	Number
Houston jobs 2010 / 2019 / 2023	2.53M / 3.13M / 3.35M
2019-23 job growth: Houston vs Dallas-Fort Worth vs Austin	6.9% vs 12.3% vs 18.0%
Mining jobs 2014 to 2023	116,000 to 75,000 (-35%)
Harris County vs Montgomery growth since 2019	+4.5% vs +20.5%
Commutes of 25+ miles, 2015 to 2023	22.5% to 26.9%
Priority tracts and workers	93 tracts, 49,704 workers (9.8%)

Likely questions

- **Why tracts instead of blocks?** Blocks contain privacy noise and are too detailed; tracts keep the totals exact and are easier to map and join to other Census data.
- **How do you know the data is correct?** Automatic checks on all 31 files, such as breakdowns adding to the total, all passed.
- **What is the biggest limitation?** Wage bands are in fixed old dollars and remote workers are counted at the employer, so I stress direction rather than exact values.
- **What next?** Add vehicle-access and poverty data, transit routes, road-network travel times, and a growth-gap breakdown versus Dallas.